VAIRAMUTHU V

Lead Engineer – QA Automation | Playwright (TypeScript & Python) | API & SQL Testing

Chennai, India | +91 7022361067 | vairamuthuv210397@gmail.com | linkedin.com/in/s

PROFESSIONAL SUMMARY

Results-driven QA Automation Lead with 4.9+ years of experience architecting and scaling test automation frameworks using Playwright (TypeScript & Python), Selenium, and Robot Framework. Skilled in API test automation, SQL-based data validation, and CI/CD pipeline integration using the Page Object Model (POM). Proven record of leading automation modernization initiatives, mentoring QA teams, and delivering reliable, high-coverage test solutions across web and embedded systems in Agile environments.

CORE SKILLS

Automation Tools: Playwright (TypeScript & Python), Selenium WebDriver, Robot Framework, Java

API Testing: API Automation Testing, Postman, Playwright API Testing

Database & Query: SQL (Data Validation, Backend Testing, Query Writing)

Languages & Scripting: TypeScript, Python, SQL, Robot Framework

Framework Design: Page Object Model (POM), Reusable & Scalable Automation Architecture, BDD Concepts

CI/CD & DevOps: Jenkins, CI/CD Pipeline Integration, Git-based Version Control

Version Control: Git, Bitbucket, Accurev

Project Management: JIRA, Agile/Scrum Methodology, Test Planning & Reporting

Testing Types: Smoke, Sanity, Regression, UI, API, Embedded LUI, Cross-Browser, Manual Testing

Operating Systems: Windows, Linux, Terminal Operations

PROFESSIONAL EXPERIENCE

Lead Engineer — HCL Tech *Mar 2023 – Present*

- Driving adoption of Playwright (TypeScript) alongside the existing Python/Robot Framework stack to modernize UI automation.
- Automated 800 of 2,500 manual test cases (32% automation coverage), significantly reducing manual regression effort and freeing up team capacity for exploratory and new-feature testing.
- Developed and maintained automation scripts using Playwright (TypeScript & Python) and Robot Framework for UI, API, and web testing, including API automation and status validation with Postman and Playwright API.
- Architected a scalable Page Object Model (POM)-based automation framework, improving script reusability and maintainability across the 800-test-case automated suite.
- Performed SQL-based backend data validation to verify data integrity between UI actions and database records.
- Integrated automated regression suite into Jenkins CI/CD pipelines, enabling faster, more frequent regression runs and expanding overall test coverage beyond what manual cycles could achieve.
- Managed code versioning using Git and performed Linux-based device interactions via terminal for embedded and web testing.
- Owned defect lifecycle, test reporting, and traceability using JIRA in an Agile/Scrum environment; mentored the QA team on automation best practices.
- Automated Web UI and embedded system test cases using Robot Framework and Python, building a reusable component library.
- Built reusable automation modules and POM layers to improve maintainability and reduce test development time.
- Executed Smoke, Sanity, and Regression testing cycles; documented results in Requirement Traceability Matrix (RTM).
- Used JIRA for bug tracking, requirement traceability, and comprehensive test documentation across releases.

KEY PROJECTS

Xerox Automation *(1759 Days)*

- Automation tester for Xerox projects, focusing on embedded device and web-based automation with Robot Framework.

ACHIEVEMENTS

- Led migration and adoption of Playwright (TypeScript) to modernize UI automation, improving execution speed and cross-browser coverage.
- Introduced API (Postman, Playwright API) and SQL-based validation layers into the existing automation framework, strengthening end-to-end test reliability.

EDUCATION

Diploma Electrical and Electronics Engineering — Sri Vasta Polytechnic College, Metamalai
2018 | Grade: 7.5/10

LANGUAGES

English, Tamil, Telugu, Kanada